

Technical Document #465: Cybersecurity - Comprehensive Overview

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Zero trust architecture assumes no implicit trust and verifies every request. It minimizes the attack surface and limits lateral movement.

Network segmentation divides networks into smaller segments. It limits the spread of attacks and improves security monitoring.

Encryption converts plaintext into ciphertext to protect data confidentiality. AES and RSA are widely used encryption algorithms.

Incident response plans define how organizations respond to security incidents. They include preparation, detection, containment, and recovery phases.

Penetration testing simulates cyber attacks to identify vulnerabilities. It helps organizations assess their security posture before real attacks occur.

Vulnerability management identifies, classifies, and remediates security vulnerabilities. Regular scanning and patching are essential components.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

The Internet of Things (IoT) connects billions of devices worldwide. This connectivity generates massive amounts of data that can be analyzed for insights.

Key Takeaways:

- Security information and event management (SIEM) systems collect and analyze security logs.
- Social engineering manipulates people into revealing confidential information.
- Multi-factor authentication (MFA) requires multiple forms of verification.